

Figure 2: **LeJEPA Theory Illustration. (left)** The world has clean latent structure (Gaussian, disentangled) with correlated positive pairs. **(center)** An unknown nonlinear process produces the observations we actually see, scrambling the latent structure. **(right)** LeJEPA trains a representation with two objectives: pull positive pairs together (*attract*) and keep the embedding distribution Gaussian (SIGReg) to prevent collapse. We prove that the learned representation must be a rotation of the true latents, i.e., the representation is forced to learn the correct World Model (Thm. 1).